

# Project Report: AI Call Agent for CognitBotz

Date: November 19, 2025

Status: In Progress

Subject: Technical Implementation & Architecture Review

## 1. Executive Summary

The objective of this project is to develop a low-latency, real-time AI Voice Agent for CognitBotz. The system handles telephony via Twilio, answering complex technical questions based on company documentation (RFPs, brochures) while maintaining natural conversational flow with barge-in capabilities.

## 2. Solution Architecture

To meet the requirements for high concurrency and low latency, the backend is built on **Python** using **FastAPI** and **WebSockets** for real-time audio orchestration.

### Core Components

The architecture mimics biological processing centers:

- **Telephony (Interface):** Twilio Media Streams handling bi-directional audio.
- **LLM (Brain):** Groq API running **Llama 3** for ultra-fast inference speeds.
- **STT (Ears):** Groq Whisper for speech-to-text transcription.
- **TTS (Mouth):** Piper (Local ONNX model) selected for near-zero latency speech synthesis.

## 3. Key Features Implementation

### 3.1 Advanced Retrieval-Augmented Generation (RAG)

The system allows the agent to consult internal documents before answering.

- **Multi-Doc Ingestion:** Automated loading of PDF documents from the data/ directory.
- **Smart Scoring Algorithm:** A custom ranking system prioritizes documents containing key technical terms (e.g., "requirements", "scope") and specific filenames to improve relevance.
- **Persistence:** The vector index (faiss\_index) is serialized to disk, preventing the need to rebuild the knowledge base on system restarts.
- **Hallucination Control:** Strict system prompting ensures the agent cites sources and adheres to facts contained in the retrieved context.

### 3.2 Robust Voice Activity Detection (VAD)

To ensure the agent feels "human," it must handle interruptions gracefully.

- **Silero VAD:** Integration of the industry-standard Silero model for precise speech detection.
- **Barge-In Logic:** The agent actively listens while speaking. Upon detecting user speech, it triggers an immediate cancellation of the current audio stream and clears the buffer, allowing for seamless interruptions.

## 4. Project Status

| Component                   | Status                                        | Notes                          |
|-----------------------------|-----------------------------------------------|--------------------------------|
| Server & Twilio Integration | <input checked="" type="checkbox"/> Completed | WebSocket orchestration active |
| RAG Knowledge Base          | <input checked="" type="checkbox"/> Completed | Optimized with smart scoring   |
| VAD & Interruptions         | <input checked="" type="checkbox"/> Completed | Barge-in logic verified        |
| Frontend Dashboard          | <input type="checkbox"/> Pending              | Next phase of development      |

*Report generated based on project notes.*